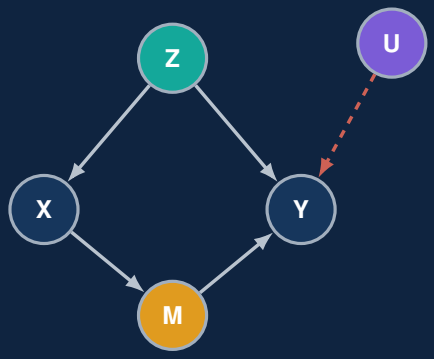


ADVANCED ARTIFICIAL INTELLIGENCE

Probabilistic Graphical Models and Causal Inference

Session 10

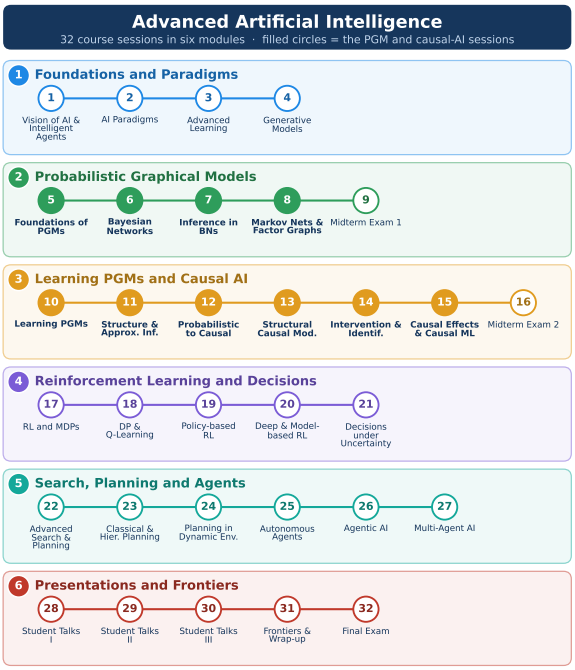
Learning: Parameter Estimation and Latent Variables

Lecture Notes

Instructor: Prof. M. Arashi

About these notes

These notes cover Session 10 of the course *Advanced Artificial Intelligence*, part of its probabilistic-graphical-model and causal-AI block (course sessions 5–15). The session follows a fixed structure: a roadmap and learning objectives, the core theory organized in parts, worked numerical examples, and a part connecting the topic to intelligent agents, agentic AI, reinforcement learning, multi-agent systems and the other sessions of the course. Every number is reproduced by the companion Python scripts (`python/s10_examples.py` for the numbers, `python/s10_figures.py` for the figures), and `questions/s10_questions.tex` contains ten questions with complete answers.



The Advanced Artificial Intelligence course. Filled circles mark the PGM and causal-AI sessions; this is session 10.

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SESSION 10 ADVANCED ARTIFICIAL INTELLIGENCE

Learning: Parameter Estimation and Latent Variables

Parameter learning, maximum likelihood, Bayesian learning and latent variables

Advanced Artificial Intelligence · Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

PART I
Maximum Likelihood
Decomposable likelihood, counts as sufficient statistics, cross-entropy.

PART II
Bayesian Learning
Beta and Dirichlet priors, smoothing, parameter independence.

PART III
Latent Variables & EM
Missing data, ELBO, EM for mixtures and for BNs.

PART IV
AI Connections
LLMs, VAEs, model-based RL, bandits, agents, games.

LEARNING OBJECTIVES

- Derive the MLE of every CPT from the decomposable log-likelihood; identify family counts as sufficient statistics.
- Learn with Beta/Dirichlet priors: posterior, predictive, MAP, smoothing, parameter independence, online updates.
- Classify missing data as MCAR, MAR or MNAR and decide when the mechanism can be ignored.
- State, prove and run EM: the ELBO, Gaussian mixtures, expected counts in Bayesian networks.
- Quantify the precision of a learned CPT entry and the benefit of shrinkage.
- Recognise these estimators in LLMs, VAEs, model-based RL, Thompson sampling, tool-using agents and games.

Where we are. Sessions 5–8 (course sessions 5–8) built the *representation* and *reasoning* (exact inference) halves of PGMs; course session 9 was the first midterm. Today we add the third pillar, *learning*: where do the numbers in the CPTs come from? Session 11 then learns the *graph itself* and replaces exact inference by sampling, which is also what the E-step of today's EM algorithm needs when the network is large.

OPENING MOTIVATION

Every model in Sessions 6–8 assumed the numbers were simply handed to us: the student network's $P(\text{Grade} \mid \text{Difficulty}, \text{Intelligence})$, clique potentials, and so on. In reality, these

numbers come from data. If we are lucky, every variable in our graph is observed in every data record, and estimating parameters is almost mechanical — a direct payoff of the factorization we spent two sessions establishing. But often the most useful variables in a model, like a customer's underlying "segment" or a patient's true (versus tested) disease status, are never directly observed. Today we handle both cases: maximum likelihood estimation when everything is observed, and the EM algorithm — built directly on Session 7's inference machinery — when it is not.

PART I

Maximum Likelihood with Complete Data

Learning a Bayesian network is counting

10.1 The Learning Problem

Notation $\mathcal{D} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)}\}$: data $\quad \mathrm{pa}_i = \mathrm{pa}(X_i)$ $\quad u$: a value of pa_i $\quad \theta_{x|u} = P(X_i{=}x \mid \mathrm{pa}_i{=}u)$ $\quad N[x,u]$, $N[u]$: counts

Suppose we are given the DAG G of a Bayesian network over $X_1, \dots, X_n$ and a dataset $\mathcal{D} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)}\}$ of N fully-observed i.i.d. samples. We want $\hat{\theta} = \arg\max_\theta \; P(\mathcal{D} \mid \theta)$. The *likelihood* $L(\theta; \mathcal{D}) = \prod_j P(\mathbf{x}^{(j)}; \theta)$ is the probability of the observed data viewed as a function of the parameters, and $\ell = \log L$ is the log-likelihood. The i.i.d. assumption is itself a conditional-independence statement: the records are independent *given* θ (Session 5).

10.2 The Decomposable Likelihood and Sufficient Statistics

Proposition 10.1 (Decomposable log-likelihood)**.** *For a Bayesian network with parameters $\theta = (\theta_1, \dots, \theta_n)$, where θ_i parameterizes $P(x_i \mid \mathrm{pa}(x_i); \theta_i)$, the log-likelihood decomposes as*

$$\ell(\theta; \mathcal{D}) = \log \prod_{j=1}^{N} \prod_{i=1}^{n} P\big(x_i^{(j)} \mid \mathrm{pa}(x_i)^{(j)}; \theta_i\big) = \sum_{i=1}^{n} \underbrace{\sum_{j=1}^{N} \log P\big(x_i^{(j)} \mid \mathrm{pa}(x_i)^{(j)}; \theta_i\big)}_{\ell_i(\theta_i; \mathcal{D})} .$$

This is an immediate consequence of the DAG factorization from Session 6, combined with log turning products into sums. Its significance is entirely practical: if the parameter sets $\theta_1, \dots, \theta_n$ do not overlap (the usual case), *maximizing $\ell(\theta; \mathcal{D})$ decomposes into n separate, independent optimization problems*, one per node, each involving only that node and its parents.

Consequence: closed-form CPT estimates. For discrete CPTs, each $\ell_i(\theta_i)$ is maximized,

subject to each row of the CPT summing to one, at the empirical conditional frequency

$$\hat{P}(X_i = v \mid \mathrm{pa}_i = u) = \frac{\mathrm{count}(X_i{=}v,\ \mathrm{pa}_i{=}u)}{\mathrm{count}(\mathrm{pa}_i{=}u)} = \frac{N[v,u]}{N[u]}.$$

The proof is Theorem 10.1 below (one line via Gibbs' inequality, plus the Lagrange-multiplier derivation).

INTUITION

Learning a Bayesian network with full observability reduces to counting: for every node, tally how often each value occurred alongside each parent configuration, and normalize. This is only possible *because* the graph told us, via the local Markov property, exactly which variables are relevant to each CPT — a smaller graph means fewer counts to accumulate and less data needed per parameter, directly mirroring Session 5's compactness argument.

For discrete CPTs, grouping the records by family value gives the *multinomial form* of the likelihood

$$L(\theta; \mathcal{D}) = \prod_{i=1}^{n} \prod_{u \in \mathrm{Val}(\mathrm{pa}_i)} \prod_{x \in \mathrm{Val}(X_i)} \theta_{x|u}^{N[x,u]}, \qquad N[x,u] = \sum_{j=1}^{N} \mathbb{1}\{x_i^{(j)} = x,\ \mathrm{pa}_i^{(j)} = u\}. \tag{10.1}$$

Each CPT *row* $\theta_{\cdot|u}$ appears in its own factor: the likelihood decomposes not only over nodes but also over parent configurations ("local decomposability"). For binary $A \to B$ with $\theta_0 = P(B{=}1 \mid A{=}0)$ and $\theta_1 = P(B{=}1 \mid A{=}1)$, the log-likelihood of $P(B \mid A)$ is $\ell_0(\theta_0) + \ell_1(\theta_1)$ with $\ell_a(\theta_a) = N[1,a]\log\theta_a + N[0,a]\log(1-\theta_a)$: a sum of two one-dimensional functions with no cross term (its Hessian is diagonal), so each row is maximized on its own. In short, all three learning regimes of this session (complete data, priors, hidden data) reduce to *(expected) family counts.*

Proposition 10.2 (Family counts are sufficient)**.** *The collection of family counts* $T(\mathcal{D}) = \{N[x,u]\}_{i,x,u}$ *is a sufficient statistic for* θ*; in fact it is minimal sufficient when all CPT rows are free.*

Proof. By (10.1), $P(\mathcal{D};\theta) = g(T(\mathcal{D}),\theta)\,h(\mathcal{D})$ with $h \equiv 1$, so the Fisher–Neyman factorization theorem gives sufficiency. Minimality: for two datasets with counts N and N', $P(\mathcal{D};\theta)/P(\mathcal{D}';\theta) = \prod_{i,u}\prod_x \theta_{x|u}^{d_{x,u}}$ with $d_{x,u} = N[x,u] - N'[x,u]$. The rows are free and vary independently, so the ratio is free of θ only if every row factor $\prod_x \theta_{x|u}^{d_{x,u}}$ is constant on the open simplex. Taking logs, $\sum_x d_{x,u}\log\theta_{x|u}$ constant there forces all $d_{x,u} = 0$ (e.g. for a binary row, $d_1\log\theta + d_0\log(1-\theta)$ is constant in $\theta \in (0,1)$ only if $d_1 = d_0 = 0$). Hence the ratio is free of θ iff $N = N'$. □

Theorem 10.1 (MLE of a CPT)**.** *If* $N[u] = \sum_x N[x,u] > 0$*, the unique maximizer of* ℓ *over the probability simplex of row* u *is* $\hat{\theta}_{x|u} = N[x,u]/N[u]$*.*

Proof. The row contributes $\ell_u(\theta) = \sum_x N[x,u]\log\theta_{x|u} = N[u]\sum_x \hat{p}_x \log\theta_{x|u}$ with $\hat{p}_x = N[x,u]/N[u]$. By Gibbs' inequality, $\sum_x \hat{p}_x\log\theta_x \le \sum_x \hat{p}_x\log\hat{p}_x$ with equality iff $\theta = \hat{p}$, because the difference is $-\mathrm{KL}(\hat{p}\,\|\,\theta) \le 0$.

Same result by a Lagrange multiplier. Maximize $\sum_x N[x,u]\log\theta_x + \lambda\big(1 - \sum_x \theta_x\big)$. Setting the derivative in θ_x to zero gives $N[x,u]/\theta_x = \lambda$, i.e. $\theta_x = N[x,u]/\lambda$; summing over x and using $\sum_x \theta_x = 1$ gives $\lambda = \sum_x N[x,u] = N[u]$, hence $\theta_x = N[x,u]/N[u]$. □

Worked Example 10.1 (Counting CPTs for the student network)**.** Suppose 200 fully-observed student records give: among the 80 students with difficulty $D = \text{hard}$, 32 have intelligence $I = \text{high}$; among the 120 with $D = \text{easy}$, 60 have $I = \text{high}$. In the student network of Session 6, D and I are root nodes (no parents, hence marginally independent), so their CPTs are the marginals $P(D)$ and $P(I)$, not $P(I \mid D)$. Counting gives $\hat{P}(D{=}\text{hard}) = 80/200 = 0.40$ and $\hat{P}(I{=}\text{high}) = (32 + 60)/200 = 92/200 = 0.46$. For the grade CPT $P(\text{Grade} \mid D, I)$ we would similarly tally grade outcomes within each of the four (D, I) combinations and normalize each of those four rows separately — exactly Proposition 10.1's claim that each CPT is estimated independently of every other one.

PYTHON LAB · `s10_examples.py` · `ex_counting_student()`

Prints `P_hat(I=high) = (32+60)/200 = 0.46` and `P_hat(D=hard) = 80/200 = 0.40`.

Remark 10.1 (Parameter sharing). In an HMM the transition CPT is shared by all time steps; counts are simply *pooled*, $N[s', s] = \sum_t \mathbb{1}\{s_{t+1} = s', s_t = s\}$. When the shared CPD is a neural network (an LLM, Part V) the closed form is lost and we use gradient ascent.

PART I · MAXIMUM LIKELIHOOD

10.3 Maximum Likelihood as Cross-Entropy Minimization

Proposition 10.3 (MLE minimizes KL to the empirical distribution)**.** *Let $\hat{p}_{\mathcal{D}}$ be the empirical distribution of the data. Then $\frac{1}{N}\ell(\theta; \mathcal{D}) = \sum_{\mathbf{x}} \hat{p}_{\mathcal{D}}(\mathbf{x}) \log p_\theta(\mathbf{x}) = -H(\hat{p}_{\mathcal{D}}) - \mathrm{KL}(\hat{p}_{\mathcal{D}} \,\|\, p_\theta)$, so $\arg\max_\theta \ell = \arg\min_\theta \mathrm{KL}(\hat{p}_{\mathcal{D}} \| p_\theta) = \arg\min_\theta H(\hat{p}_{\mathcal{D}}, p_\theta)$ (cross-entropy).*

Proof. Group equal records: $\frac{1}{N}\sum_j \log p_\theta(\mathbf{x}^{(j)}) = \sum_{\mathbf{x}} \hat{p}(\mathbf{x}) \log p_\theta(\mathbf{x})$, then add and subtract $\sum \hat{p} \log \hat{p}$. □

COMMON PITFALL A zero count gives $\hat{\theta}_{x|u} = 0$: any future record containing (x, u) gets log-likelihood $-\infty$. With many parent configurations zero counts are the rule. Part II fixes this with priors.

PART II

Bayesian Parameter Learning

Parameters are random variables too

PART II · BAYESIAN LEARNING

10.4 The Beta–Bernoulli Model

In the Bayesian view θ is itself a random variable with a prior $p(\theta)$, and learning is inference: $p(\theta \mid \mathcal{D}) \propto p(\theta)L(\theta;\mathcal{D})$. Graphically, θ becomes a parent of every record, and the records are conditionally independent given θ but *not* marginally independent: each record carries information about the next.

Theorem 10.2 (Beta–Bernoulli conjugacy)**.** *Let* $\theta \sim \mathrm{Beta}(\alpha,\beta)$ *and* $X_1,\dots,X_n \mid \theta \overset{iid}{\sim} \mathrm{Bernoulli}(\theta)$ *with* $k=\sum_i X_i$*. Then*

(a) $\theta \mid \mathcal{D} \sim \mathrm{Beta}(\alpha+k,\ \beta+n-k)$*;*

(b) posterior predictive: $P(X_{n+1}=1 \mid \mathcal{D}) = \mathbb{E}[\theta \mid \mathcal{D}] = \dfrac{\alpha+k}{\alpha+\beta+n}$*;*

(c) posterior mean $= w_n\dfrac{k}{n} + (1-w_n)\dfrac{\alpha}{\alpha+\beta}$ *with* $w_n = \dfrac{n}{n+\alpha+\beta}$*; MAP* $= \dfrac{\alpha+k-1}{\alpha+\beta+n-2}$ *when* $\alpha+k>1,\ \beta+n-k>1$*.*

Proof. (a) $p(\theta \mid \mathcal{D}) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}\cdot\theta^{k}(1-\theta)^{n-k}$, the kernel of $\mathrm{Beta}(\alpha+k,\beta+n-k)$. (b) $P(X_{n+1}{=}1 \mid \mathcal{D}) = \int \theta\, p(\theta \mid \mathcal{D})\,d\theta$ because $X_{n+1} \perp\!\!\!\perp \mathcal{D} \mid \theta$; the mean of $\mathrm{Beta}(a,b)$ is $a/(a+b)$. (c) Split the numerator of (b): $\frac{\alpha+k}{\alpha+\beta+n} = \frac{n}{\alpha+\beta+n}\cdot\frac{k}{n} + \frac{\alpha+\beta}{\alpha+\beta+n}\cdot\frac{\alpha}{\alpha+\beta} = w_n\frac{k}{n} + (1{-}w_n)\frac{\alpha}{\alpha+\beta}$. The MAP is the mode $(a{-}1)/(a{+}b{-}2)$ of $\mathrm{Beta}(a,b)$, obtained by setting the derivative of $(a{-}1)\log\theta + (b{-}1)\log(1{-}\theta)$ to zero. □

The prior acts as $\alpha+\beta$ *pseudo-observations*, the equivalent sample size: the posterior mean is a convex combination of the MLE and the prior mean, with weight $w_n \to 1$ as data accumulate.

Proposition 10.4 (Sequential = batch updating)**.** *If records are conditionally i.i.d. given* θ*, then for any split* $\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2$*,* $p(\theta \mid \mathcal{D}) \propto p(\theta \mid \mathcal{D}_1)\,P(\mathcal{D}_2 \mid \theta)$*: yesterday's posterior is today's prior.*

Proof. $p(\theta \mid \mathcal{D}) \propto p(\theta)P(\mathcal{D}_1 \mid \theta)P(\mathcal{D}_2 \mid \theta) \propto p(\theta \mid \mathcal{D}_1)P(\mathcal{D}_2 \mid \theta)$, using $\mathcal{D}_1 \perp\!\!\!\perp \mathcal{D}_2 \mid \theta$. □

For conjugate models this is "add the new counts to the hyperparameters": an *online* algorithm with $O(1)$ memory per CPT row.

Worked Example 10.2 (Sequential Beta updating)**.** Prior $\mathrm{Beta}(\alpha,\beta) = \mathrm{Beta}(2,2)$; data $1,1,0,1,1,1,0,1,1,1$, so $n=10$ and $k=8$.

1. *Sequential.* Each 1 adds one to the first parameter, each 0 to the second: after record 1 ($x=1$) $\mathrm{Beta}(3,2)$; after records 2–3 $(1,0)$ $\mathrm{Beta}(4,3)$; after all ten, $\mathrm{Beta}(2+8,\ 2+2) = \mathrm{Beta}(10,4)$, the same as the batch update of Theorem 10.2(a), as Proposition 10.4 guarantees.
2. *Point estimates.* Posterior mean $10/14 = 0.7143$; MAP $(10{-}1)/(10{+}4{-}2) = 9/12 = 0.75$; MLE $k/n = 8/10 = 0.8$.
3. *Shrinkage form.* $w_n = n/(n+\alpha+\beta) = 10/14 = 0.7143$ and the prior mean is $2/4 = 0.5$: $0.7143 \times 0.8 + 0.2857 \times 0.5 = 0.5714 + 0.1429 = 0.7143$.

4. *Credible interval.* The 2.5% and 97.5% quantiles of Beta(10, 4) (no closed form; computed with `scipy.stats.beta.ppf`) give the 95% equal-tailed interval (0.4619, 0.9091).

PYTHON LAB · `s10_examples.py` · `ex_beta_sequence()`

`after 1 obs: Beta(3,2), after 3 obs: Beta(4,3), after 10 obs: Beta(10,4); batch update: Beta(10,4) (identical)`; mean 0.7143, MAP 0.7500, CI (0.4619, 0.9091).

PART II · BAYESIAN LEARNING

10.5 Dirichlet Priors for CPTs

Definition 10.1 (Dirichlet distribution)**.** $\theta = (\theta_1, \dots, \theta_K)$ on the simplex has a $\mathrm{Dir}(\alpha_1, \dots, \alpha_K)$ distribution, $\alpha_k > 0$, if $p(\theta) = \frac{\Gamma(\alpha_0)}{\prod_k \Gamma(\alpha_k)} \prod_k \theta_k^{\alpha_k - 1}$, $\alpha_0 = \sum_k \alpha_k$. Its mean is α_k/α_0 and for $K = 2$ it is the Beta distribution.

Theorem 10.3 (Dirichlet–multinomial conjugacy)**.** *If $\theta \sim \mathrm{Dir}(\boldsymbol{\alpha})$ and the counts are $\mathbf{N} = (N_1, \dots, N_K)$, $N = \sum_k N_k$, then $\theta \mid \mathcal{D} \sim \mathrm{Dir}(\boldsymbol{\alpha} + \mathbf{N})$, the predictive probability of category k is $(\alpha_k + N_k)/(\alpha_0 + N)$, and the MAP is $(\alpha_k + N_k - 1)/(\alpha_0 + N - K)$ whenever all $\alpha_k + N_k > 1$.*

Proof. Prior $\times$ likelihood $\propto \prod_k \theta_k^{\alpha_k - 1} \prod_k \theta_k^{N_k} = \prod_k \theta_k^{\alpha_k + N_k - 1}$. The predictive is the posterior mean. For the mode, maximize $\sum_k (a_k - 1) \log \theta_k$ on the simplex with a Lagrange multiplier as in Theorem 10.1, with $a_k - 1$ in place of $N[x, u]$. □

Laplace smoothing is a Dirichlet prior. With $\alpha_k = 1$ (uniform prior) the predictive is $(N_k + 1)/(N + K)$, Laplace's "add-one" rule; $\alpha_k = \frac{1}{2}$ is the Jeffreys prior ("add one half"). Note the subtle identity: the *MAP* under $\mathrm{Dir}(2, \dots, 2)$ equals the *posterior mean* under $\mathrm{Dir}(1, \dots, 1)$.

Worked Example 10.3 (Smoothing a CPT row with a zero count)**.** A diagnostic network records the symptom level (none, mild, severe) of 10 patients with a given disease: counts $(6, 4, 0)$. Here $N = 10$ and $K = 3$. The MLE $(6, 4, 0)/10 = (0.6, 0.4, 0)$ declares a severe symptom impossible. By Theorem 10.3:

1. $\mathrm{Dir}(1, 1, 1)$, $\alpha_0 = 3$: posterior mean $(6 + 1, 4 + 1, 0 + 1)/(10 + 3) = (7, 5, 1)/13 = (0.5385, 0.3846, 0.0769)$.
2. Jeffreys $\mathrm{Dir}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, $\alpha_0 = 1.5$: posterior mean $(6.5, 4.5, 0.5)/11.5 = (0.5652, 0.3913, 0.0435)$.
3. MAP under $\mathrm{Dir}(2, 2, 2)$, $\alpha_0 = 6$: $(6+2-1, 4+2-1, 0+2-1)/(10+6-3) = (7, 5, 1)/13$, again $(0.5385, 0.3846, 0.0769)$.

PYTHON LAB · `s10_examples.py` · `ex_dirichlet_smoothing()`

`MLE [0.6 0.4 0.]; posterior mean Dir(1.0) [0.5385 0.3846 0.0769]; Dir(0.5) [0.5652 0.3913 0.0435]; MAP under Dir(2,2,2) [0.5385 0.3846 0.0769].`

10.5.1 Parameter independence and the Bayesian network as a whole

Definition 10.2 (Global and local parameter independence)**.** A prior over the parameters of a BN satisfies *global parameter independence* if $p(\theta) = \prod_i p(\theta_{X_i|\mathrm{pa}_i})$, and *local parameter independence* if moreover $p(\theta_{X_i|\mathrm{pa}_i}) = \prod_u p(\theta_{X_i|u})$ (CPT rows independent). This is the standard assumption of Heckerman, Geiger and Chickering (1995).

Theorem 10.4 (Posterior parameter independence)**.** *Under global and local parameter independence with Dirichlet priors $\theta_{X_i|u} \sim \mathrm{Dir}(\boldsymbol{\alpha}_{i,u})$ and complete data, the posterior factorizes as $p(\theta \mid \mathcal{D}) = \prod_i \prod_u \mathrm{Dir}\big(\theta_{X_i|u};\ \boldsymbol{\alpha}_{i,u} + \mathbf{N}_{i,u}\big)$, where $\mathbf{N}_{i,u} = (N[x,u])_x$.*

Proof. By (10.1) the likelihood is a product of factors $\prod_x \theta_{x|u}^{N[x,u]}$, one for each pair (i,u), and by assumption so is the prior. The product of the two is therefore $\prod_{i,u}\big[p(\theta_{X_i|u}) \prod_x \theta_{x|u}^{N[x,u]}\big]$; each bracket depends on one block only, so the blocks are independent a posteriori, and each bracket is the Dirichlet kernel of Theorem 10.3. □

Proposition 10.5 (Bayesian prediction = plug in posterior means)**.** *Under the conditions of Theorem 10.4, $P(\mathbf{x}^{(N+1)} \mid \mathcal{D}) = \prod_i \dfrac{\alpha_{x_i|u_i} + N[x_i,u_i]}{\alpha_{u_i} + N[u_i]}$, where u_i is the parent configuration of X_i in the new record and $\alpha_u = \sum_x \alpha_{x|u}$.*

Proof. $P(\mathbf{x}^{(N+1)} \mid \mathcal{D}) = \int \prod_i \theta_{x_i|u_i}\, p(\theta \mid \mathcal{D})\, d\theta$. Each factor involves a different block (i,u_i) and the blocks are independent, so the integral is the product of the posterior means $\mathbb{E}[\theta_{x_i|u_i} \mid \mathcal{D}]$. □

Worked Example 10.4 (Bayesian prediction in $A \to B$)**.** Take $N = 10$ complete records $(A,B) = (1,1) \times 4,\ (1,0) \times 1,\ (0,1) \times 2,\ (0,0) \times 3$ (MLEs $\hat{P}(A{=}1) = 5/10 = 0.5$, $\hat{P}(B{=}1 \mid A{=}1) = 4/5 = 0.8$, $\hat{P}(B{=}1 \mid A{=}0) = 2/5 = 0.4$), and independent $\mathrm{Dir}(1,1)$ (uniform) priors on θ_A, $\theta_{B|a^1}$ and $\theta_{B|a^0}$. The counts are $N[A{=}1] = 5$, $N[A{=}0] = 5$ and $N[B{=}1, A{=}1] = 4$. By Proposition 10.5,

$$P(A{=}1 \mid \mathcal{D}) = \frac{1+5}{2+10} = \frac{6}{12} = 0.5000, \qquad P(B{=}1 \mid A{=}1, \mathcal{D}) = \frac{1+4}{2+5} = \frac{5}{7} = 0.7143,$$

so $P(A{=}1, B{=}1 \mid \mathcal{D}) = 0.5000 \times 0.7143 = 0.3571$, slightly below the plug-in MLE value $0.5 \times 0.8 = 0.4$.

PYTHON LAB · `s10_examples.py` · `ex_bayes_prediction_bn()`

```
P(A=1 | D) = 6/12 = 0.5000; P(B=1 | A=1, D) = 5/7 = 0.7143; P(A=1, B=1 | D) = 0.3571.
```

PART III

Missing Data, Latent Variables and EM

When the log of a sum refuses to split

PART III · LATENT VARIABLES

10.6 Missing Data Mechanisms

Let $\mathbf{Y}$ be the full data, split into observed and missing parts $(\mathbf{Y}_{\text{obs}}, \mathbf{Y}_{\text{mis}})$, and let $\mathbf{R}$ be the vector of missingness indicators ($R = 1$ if the entry is missing). The data model is $p(\mathbf{y} \mid \theta)$ and the *missingness mechanism* is $p(\mathbf{r} \mid \mathbf{y}, \psi)$ (Rubin, 1976; Little and Rubin, 2002).

Definition 10.3 (MCAR, MAR, MNAR)**.** The data are *missing completely at random* (MCAR) if $p(\mathbf{r} \mid \mathbf{y}, \psi) = p(\mathbf{r} \mid \psi)$; *missing at random* (MAR) if $p(\mathbf{r} \mid \mathbf{y}, \psi) = p(\mathbf{r} \mid \mathbf{y}_{\text{obs}}, \psi)$; and *missing not at random* (MNAR) otherwise. In graphical terms: MCAR means R has no parents among the variables, MAR means $R \perp\!\!\!\perp \mathbf{Y}_{\text{mis}} \mid \mathbf{Y}_{\text{obs}}$, MNAR means R depends on the value that is missing.

Theorem 10.5 (Ignorability under MAR)**.** *If the data are MAR and θ and ψ are variation independent (distinct), then the likelihood of the observed data factorizes as $L(\theta, \psi; \mathbf{y}_{obs}, \mathbf{r}) = p(\mathbf{r} \mid \mathbf{y}_{obs}, \psi) \cdot p(\mathbf{y}_{obs} \mid \theta)$; hence the MLE of θ (and the posterior of θ under independent priors) can be computed from $p(\mathbf{y}_{obs} \mid \theta) = \int p(\mathbf{y}_{obs}, \mathbf{y}_{mis} \mid \theta)\, d\mathbf{y}_{mis}$ alone.*

Proof. $L = \int p(\mathbf{y}_{\text{obs}}, \mathbf{y}_{\text{mis}} \mid \theta)\, p(\mathbf{r} \mid \mathbf{y}_{\text{obs}}, \mathbf{y}_{\text{mis}}, \psi)\, d\mathbf{y}_{\text{mis}}$. Under MAR the second factor does not depend on $\mathbf{y}_{\text{mis}}$ and leaves the integral; the remaining integral is $p(\mathbf{y}_{\text{obs}} \mid \theta)$. The maximization over (θ, ψ) then separates because the parameter space is a product. □

PART III · LATENT VARIABLES

10.7 When Observability Fails: Latent Variables

Now suppose some variables are never recorded. A marketing team observes purchase amounts but never directly asks customers which underlying "segment" (budget-conscious, aspirational, luxury) they belong to; a language model sees words but never sees the latent topic that generated them. Call the observed variables X and the latent (hidden) variables C. In *plate notation* a box around the per-record variables marks N i.i.d. copies, a compact way to draw "repeat this substructure N times". A latent variable is the extreme case of missing data: it is missing in every record, so $R \equiv 1$ does not depend on any value; the mechanism is trivially MCAR (hence MAR), and Theorem 10.5 lets us work with $p(\mathbf{y}_{\text{obs}} \mid \theta)$ alone.

The problem: Proposition 10.1's decomposition assumed we could plug $\mathbf{x}^{(j)}$ directly into every factor. With C unobserved, the likelihood of the data we *can* see requires summing C out:

$$\ell(\theta; \mathcal{D}) = \sum_{j=1}^{N} \log P(x^{(j)}; \theta) = \sum_{j=1}^{N} \log \sum_{c} P(x^{(j)}, c; \theta),$$

and the log of a *sum* no longer decomposes into separate per-node terms the way Proposition 10.1 did. This is not a minor inconvenience — it usually makes direct maximization analytically intractable.

PART III · LATENT VARIABLES

10.8 The EM Algorithm

The Expectation-Maximization (EM) algorithm (Dempster, Laird and Rubin, 1977) sidesteps this by alternating between (1) using the *current* parameter estimate to infer a distribution over the missing C given what is observed — exactly a Session-3 inference query — and (2) treating those inferred (soft) values as if they were observed, reducing back to the easy, decomposable MLE problem of Part I.

Algorithm 10.1 Expectation–Maximization

1: Initialize $\theta^{(0)}$ arbitrarily (e.g., randomly).
2: **for** $t = 0, 1, 2, \ldots$ until convergence **do**
3: **E-step:** for each data point j, compute $q_j(c) := P(c \mid x^{(j)}; \theta^{(t)})$ (an inference query — Session 7's VE).
4: **M-step:** set $\theta^{(t+1)} = \arg\max_\theta \sum_j \sum_c q_j(c) \log P(x^{(j)}, c; \theta)$ (an ordinary weighted MLE, decomposable exactly as in Proposition 10.1, now with "soft counts" $q_j(c)$ replacing observed indicator values).
5: **end for**

Proposition 10.6 (ELBO decomposition)**.** *For any distribution q over c and any θ,* $\log P(x;\theta) = \underbrace{\sum_c q(c) \log \frac{P(x,c;\theta)}{q(c)}}_{\mathcal{L}(q,\theta)\ \textit{(ELBO)}} + \mathrm{KL}\big(q \,\|\, P(\cdot \mid x;\theta)\big)$.

Proof. Write $P(x, c; \theta) = P(x; \theta)P(c \mid x; \theta)$ inside $\mathcal{L}$: $\mathcal{L}(q, \theta) = \log P(x; \theta) - \sum_c q(c) \log \frac{q(c)}{P(c|x;\theta)}$. □

For a dataset, apply Proposition 10.6 to each record with its own q_j and sum: $\ell(\theta; \mathcal{D}) = \mathcal{L}(q, \theta) + \sum_j \mathrm{KL}\big(q_j \,\|\, P(\cdot \mid x^{(j)}; \theta)\big)$ with $\mathcal{L}(q, \theta) = \sum_j \sum_c q_j(c) \log \frac{P(x^{(j)}, c; \theta)}{q_j(c)}$. Splitting the log,

$$\mathcal{L}(q, \theta) = Q(\theta \mid \theta^{(t)}) + H(q), \qquad Q(\theta \mid \theta^{(t)}) = \sum_j \sum_c q_j(c) \log P(x^{(j)}, c; \theta), \tag{10.2}$$

with the entropy term $H(q) = -\sum_j \sum_c q_j(c) \log q_j(c)$, where Q (the *expected complete-data log-likelihood*) is evaluated with $q_j = P(\cdot \mid x^{(j)}; \theta^{(t)})$. EM is *coordinate ascent on the ELBO*: the E-step maximizes $\mathcal{L}$ over q by setting $q_j = P(\cdot \mid x^{(j)}; \theta^{(t)})$, which makes every KL term zero; the M-step maximizes $\mathcal{L}$ over θ, and since $H(q)$ does not involve θ this is exactly maximizing $Q(\theta \mid \theta^{(t)})$, the M-step of Algorithm 10.1.

Theorem 10.6 (EM monotonicity)**.** *The sequence of observed-data log-likelihoods produced by EM is non-decreasing: $\ell(\theta^{(t+1)}; \mathcal{D}) \geq \ell(\theta^{(t)}; \mathcal{D})$ for every t.*

Proof. By Proposition 10.6, $\ell(\theta) = \mathcal{F}(q, \theta) + \mathrm{KL}\big(q \,\|\, P(\mathbf{h} \mid \mathbf{o}; \theta)\big) \geq \mathcal{F}(q, \theta)$ for every q, with equality for the E-step choice $q = P(\mathbf{h} \mid \mathbf{o}; \theta^{(t)})$; so $\ell(\theta^{(t)}) = \mathcal{F}(q, \theta^{(t)}) \leq \mathcal{F}(q, \theta^{(t+1)}) \leq \ell(\theta^{(t+1)})$. □ The argument uses the decomposition (10.2) and $\mathrm{KL} \geq 0$ (Gibbs' inequality, itself Jensen's inequality for the concave log) to show the M-step maximizes a lower bound on ℓ that is guaranteed to touch ℓ exactly at $\theta^{(t)}$ — so any improvement to the bound is a genuine (or at worst equal) improvement to the true log-likelihood. EM is not guaranteed to reach a global maximum (it can and does get stuck in local optima, so multiple random restarts are standard practice), but Theorem 10.6

guarantees it never gets *worse*, which is the essential property that makes it a reliable, widely-used algorithm rather than a heuristic.

PART III · LATENT VARIABLES

10.9 EM for Gaussian Mixtures in Full

Proposition 10.7 (EM updates for a Gaussian mixture)**.** *For* $p(x) = \sum_{k=1}^{K} \pi_k N(x; \mu_k, \sigma_k^2)$*, one EM iteration is*

$$E{:}\ r_{jk} = \frac{\pi_k N(x_j; \mu_k, \sigma_k^2)}{\sum_l \pi_l N(x_j; \mu_l, \sigma_l^2)}; \quad M{:}\ \pi_k = \frac{N_k}{N}, \quad \mu_k = \frac{\sum_j r_{jk} x_j}{N_k}, \quad \sigma_k^2 = \frac{\sum_j r_{jk}(x_j - \mu_k)^2}{N_k},$$

with $N_k = \sum_j r_{jk}$*. In* d *dimensions,* $\Sigma_k = \frac{1}{N_k} \sum_j r_{jk} (\mathbf{x}_j - \boldsymbol{\mu}_k)(\mathbf{x}_j - \boldsymbol{\mu}_k)^\top$*.*

Proof. $Q(\theta) = \sum_j \sum_k r_{jk} \big[\log \pi_k - \frac{1}{2} \log(2\pi\sigma_k^2) - \frac{(x_j - \mu_k)^2}{2\sigma_k^2} \big]$. The π-part $\sum_k N_k \log \pi_k$ is maximized on the simplex at $\pi_k = N_k/N$ (Theorem 10.1). $\partial Q / \partial \mu_k = \sum_j r_{jk}(x_j - \mu_k)/\sigma_k^2 = 0$ gives the weighted mean. $\partial Q / \partial \sigma_k^2 = \sum_j r_{jk} \big[-\frac{1}{2\sigma_k^2} + \frac{(x_j - \mu_k)^2}{2\sigma_k^4} \big] = 0$ gives the weighted variance (evaluated at the new μ_k). These stationary points are maxima: Q is a concave quadratic in μ_k, and as a function of $v = \sigma_k^2$ the k-th term is $-\frac{N_k}{2} \log v - \frac{S_k}{2v}$ with $S_k = \sum_j r_{jk}(x_j - \mu_k)^2 > 0$, which tends to $-\infty$ as $v \to 0$ and as $v \to \infty$ and has the single stationary point $v = S_k / N_k$. □

Worked Example 10.5 (One full EM iteration with weights, means and variances)**.** Data $x = (1, 1.5, 2, 6, 7, 7.5)$; components $k = 0, 1$ started at $\pi = (0.5, 0.5)$, $\mu = (1, 7)$, $\sigma^2 = (1, 1)$; write $\varphi(z) = e^{-z^2/2}/\sqrt{2\pi}$.

1. *E-step.* With equal weights and variances, $r_{j0} = e^{-(x_j - 1)^2/2} / \big(e^{-(x_j-1)^2/2} + e^{-(x_j-7)^2/2} \big)$. The least certain points are $x = 2$: $r_{j0} = e^{-0.5}/(e^{-0.5} + e^{-12.5}) = 0.999994$, and $x = 6$: $r_{j0} = e^{-12.5}/(e^{-12.5} + e^{-0.5}) = 0.000006$. So, to four decimals, $r_{\cdot 0} = (1, 1, 1, 0, 0, 0)$ and $N_k = (3, 3)$.

2. *M-step.* $\pi = (3/6, 3/6) = (0.5, 0.5)$; $\mu_0 = (1 + 1.5 + 2)/3 = 1.5$, $\mu_1 = (6 + 7 + 7.5)/3 = 20.5/3 = 6.8333$; $\sigma_0^2 = \big((-0.5)^2 + 0^2 + 0.5^2 \big)/3 = 0.5/3 = 0.1667$, $\sigma_1^2 = \big((6 - 6.8333)^2 + (7 - 6.8333)^2 + (7.5 - 6.8333)^2 \big)/3 = (0.6944 + 0.0278 + 0.4445)/3 = 1.1667/3 = 0.3889$.

3. *Log-likelihood* $\ell = \sum_j \log \sum_k \pi_k N(x_j; \mu_k, \sigma_k^2)$. Before: the other component contributes at most $e^{-12.5}$, so $\log p(x_j) \approx \log \big(0.5\, \varphi(d_j) \big)$ with distance $d_j = 0, 0.5, 1$ to the nearest mean, i.e. $-1.61209, -1.73709, -2.11208$, each occurring twice: $\ell^{(0)} = 2(-1.61209 - 1.73709 - 2.11208) = -10.9225$. After the M-step the per-point terms are $(-1.46617, -0.71633, -1.46612, -2.03264, -1.17563, -1.71129)$, summing to $\ell^{(1)} = -8.5682$.

PYTHON LAB · `s10_examples.py` · `ex_gmm_full_iteration()`

```
r(c=0|x=2) = 0.999994, r(c=0|x=6) = 0.000006;   pi^(1) = [0.5 0.5], mu^(1) =
[1.5 6.8333], var^(1) = [0.1667 0.3889]; log-lik:  -10.9225 -> -8.5682.
```

COMMON PITFALL The MLE of a variance is biased, $\mathbb{E}\hat{\sigma}^2 = \frac{n-1}{n}\sigma^2$ for n i.i.d. Gaussian observations; worse, the Gaussian-mixture likelihood is *unbounded*: put $\mu_k = x_j$ and let $\sigma_k \to 0$; then the term $\pi_k N(x_j; x_j, \sigma_k^2) = \pi_k/(\sqrt{2\pi}\sigma_k) \to \infty$ while the other points keep positive density from the other components, so $\ell \to +\infty$. EM can collapse a component onto one point. In practice: floor the variances, add a weak inverse-gamma (1-D) or inverse-Wishart (d-D) prior and run MAP-EM, or restart.

A 2-D illustration (simulation, `ex_gmm_2d()`). Illustrative data: 450 points from three bivariate Gaussian clusters of sizes $180, 150, 120$ (true weights $0.4, 0.333, 0.267$; means $(0,0)$, $(4,1)$, $(1.5,4)$; covariances listed in `gmm_2d_data()`, seed 7). EM for a three-component full-covariance mixture is deliberately started badly: the three means are random points among the 60 left-most data points, every covariance equals the overall sample covariance, and the weights are equal. The log-likelihood still climbs monotonically, $-2399 \to -1799 \to -1733 \to -1555$ at iterations $0, 1, 5, 40$, and the final weights $(0.395, 0.267, 0.338)$ match the true $(0.4, 0.333, 0.267)$ up to relabeling of the components.

PYTHON LAB · `s10_examples.py` · `ex_gmm_2d()`

```
N = 450; log-lik at iterations 0, 1, 5, 40:  -2399, -1799, -1733, -1555;
monotone:  True; final weights:  [0.395 0.267 0.338].
```

PART III · LATENT VARIABLES

10.10 EM for Bayesian Networks: Expected Counts

For a BN with some entries missing in some records, the M-step needs the counts $N[x,u]$, which are not observed. Replace them by *expected sufficient statistics*

$$\bar{M}_t[x,u] = \sum_{j=1}^{N} P\big(X_i{=}x,\ \mathrm{pa}_i{=}u \mid \mathbf{o}^{(j)}; \theta^{(t)}\big), \qquad \theta_{x|u}^{(t+1)} = \frac{\bar{M}_t[x,u]}{\sum_{x'} \bar{M}_t[x',u]}, \tag{10.3}$$

where $\mathbf{o}^{(j)}$ is the observed part of record j. A record in which the whole family is observed contributes the indicator $\mathbb{1}\{x_i^{(j)}{=}x,\ \mathrm{pa}_i^{(j)}{=}u\}$, exactly as in (10.1); an incomplete record contributes a *family marginal* computed by inference (Session 7: one clique-tree calibration per record gives all family marginals at once). With a $\mathrm{Dir}(\boldsymbol{\alpha}_u)$ prior on each row, MAP-EM uses the MAP formula of Theorem 10.3 with expected counts, $\theta_{x|u}^{(t+1)} = (\bar{M}_t[x,u] + \alpha_{x|u} - 1)/\sum_{x'}(\bar{M}_t[x',u] + \alpha_{x'|u} - 1)$; with $\alpha_{x|u} = 2$ this is "add one to every expected count".

Worked Example 10.6 (EM with expected counts in $A \to B$)**.** A hypothetical log of $N = 140$ records for binary $A \to B$, with parameters $\theta_A = P(A{=}1)$, $\theta_{B|1} = P(B{=}1 \mid A{=}1)$, $\theta_{B|0} = P(B{=}1 \mid A{=}0)$:

- 80 complete records: $(A,B) = (1,1){\times}30$, $(1,0){\times}10$, $(0,1){\times}15$, $(0,0){\times}25$;
- 40 records with B missing: $A{=}1$ in 20, $A{=}0$ in 20;
- 20 records with A missing: $B{=}1$ in 12, $B{=}0$ in 8.

Start at $\theta^{(0)}$: $\theta_A = \theta_{B|1} = \theta_{B|0} = 0.5$.
Observed-data log-likelihood. A complete record contributes $\log P(a,b)$, a B-missing record $\log P(a)$, an A-missing record $\log P(b) = \log \sum_a P(a)P(b \mid a)$. At $\theta^{(0)}$ these are $\log 0.25$, $\log 0.5$ and $\log(0.5 \cdot 0.5 + 0.5 \cdot 0.5) = \log 0.5$, so $\ell^{(0)} = 80 \log 0.25 + 40 \log 0.5 + 20 \log 0.5 =$

$-110.904 - 27.726 - 13.863 = -152.492$.
E-step. For an A-missing record, $P(A{=}1 \mid B{=}b) = \frac{\theta_A P(b|A{=}1)}{\theta_A P(b|A{=}1)+(1-\theta_A)P(b|A{=}0)} = \frac{0.5\cdot 0.5}{0.5\cdot 0.5+0.5\cdot 0.5} = 0.5$ for both b, so each such record is split evenly between $A{=}1$ and $A{=}0$. For a B-missing record, $P(B{=}1 \mid A{=}a) = \theta_{B|a} = 0.5$. Expected counts, term by term (complete + B-missing + A-missing):

$$\begin{aligned}
\bar{M}[A{=}1] &= (30+10) + 20 + (0.5\cdot 12 + 0.5\cdot 8) = 40+20+10 = 70,\\
\bar{M}[A{=}0] &= (15+25) + 20 + (0.5\cdot 12 + 0.5\cdot 8) = 40+20+10 = 70,\\
\bar{M}[A{=}1, B{=}1] &= 30 + 0.5\cdot 20 + 0.5\cdot 12 = 30+10+6 = 46,\\
\bar{M}[A{=}0, B{=}1] &= 15 + 0.5\cdot 20 + 0.5\cdot 12 = 15+10+6 = 31.
\end{aligned}$$

M-step. $\theta_A = 70/140 = 0.5$, $\theta_{B|1} = 46/70 = 0.6571$, $\theta_{B|0} = 31/70 = 0.4429$; now $\ell^{(1)} = -146.881$.
Iterating 200 times (to convergence) gives $\hat{\theta}_A = 0.5019$, $\hat{\theta}_{B|1} = 0.7557$, $\hat{\theta}_{B|0} = 0.3823$, with ℓ rising monotonically from -152.492 to -145.640. The complete-case estimates $30/40 = 0.750$ and $15/40 = 0.375$ differ because the A-missing records carry information about $P(B \mid A)$ (the B-missing records, being MAR given A, only inform θ_A).

PYTHON LAB · s10_examples.py · ex_em_missing_cpt()

```
l(theta^(0)) = -110.904 + -27.726 + -13.863 = -152.492;      expected counts
M[A=1]=70.00, M[A=1,B=1]=46.00, M[A=0,B=1]=31.00;  after 1 iteration:  theta
= [0.5 0.6571 0.4429];   converged (200 iterations):  0.5019, 0.7557, 0.3823;
log-lik:  -152.492 -> -146.881 -> ...  -> -145.640; monotone:  True.
```

PART III · LATENT VARIABLES

10.11 Hard EM versus Soft EM

Hard EM replaces the posterior q_j by a point mass on its mode $\hat{c}_j = \arg\max_c P(c \mid x^{(j)}; \theta^{(t)})$ and then refits θ on the completed data. For a mixture of spherical Gaussians with equal, known variances and weights, the mode is the nearest mean, so hard EM is exactly k-means. Hard EM maximizes the joint $\max_{c_{1:N},\theta} \log P(\mathbf{x}, \mathbf{c}; \theta)$, a *different* objective from $\ell(\theta)$: it needs only (cheaper) MAP inference, but its estimates are biased, and stay biased as $N \to \infty$ (inconsistent), when components overlap.

PART III · LATENT VARIABLES

10.12 How Precise Is a Learned CPT Entry?

The estimate of a CPT entry is a proportion computed from the $N[u]$ records with parent configuration u, so its precision is governed by that *effective* sample size, not by the total N.

Theorem 10.7 (Asymptotic normality of a CPT entry)**.** *Let the records be i.i.d. from a BN*

with $p_u = P(\mathrm{pa}(X) = u) > 0$, and $\hat\theta_{x|u} = N[x,u]/N[u]$. Then $\hat\theta_{x|u} \to \theta_{x|u}$ almost surely and

$$\sqrt{N}\big(\hat\theta_{x|u} - \theta_{x|u}\big) \xrightarrow{d} N\!\left(0,\ \frac{\theta_{x|u}(1-\theta_{x|u})}{p_u}\right).$$

Moreover, estimates belonging to different parent configurations $u \neq u'$ are asymptotically independent.

(This follows from the central limit theorem for the indicators of the records with $\mathrm{pa}(X) = u$ and Slutsky's lemma; the per-observation Fisher information of a Bernoulli parameter is $I(\theta) = 1/[\theta(1-\theta)]$.) Rare parent configurations (p_u small) are therefore learned slowly, which is one more reason to keep parent sets small.

Plugging $\hat\theta$ into Theorem 10.7 gives the *Wald* interval $\hat\theta \pm z\sqrt{\hat\theta(1-\hat\theta)/n}$ with $n = N[u]$. *Inverting* the score test instead, i.e. solving $|\hat\theta - \theta| \le z\sqrt{\theta(1-\theta)/n}$ for θ, gives the quadratic inequality $(1+\frac{z^2}{n})\theta^2 - (2\hat\theta + \frac{z^2}{n})\theta + \hat\theta^2 \le 0$, whose solution is the *Wilson* (1927) interval, always inside $[0,1]$:

$$\frac{\hat\theta + \frac{z^2}{2n}}{1+\frac{z^2}{n}} \ \pm\ \frac{z}{1+\frac{z^2}{n}}\sqrt{\frac{\hat\theta(1-\hat\theta)}{n} + \frac{z^2}{4n^2}}. \tag{10.4}$$

Its centre is itself a shrinkage estimator: for $z = 2$ it is $(k+2)/(n+4)$, the posterior mean under $\mathrm{Beta}(2,2)$ (the Agresti–Coull "add two successes and two failures" rule). Shrinkage also pays in mean squared error: for $k \sim \mathrm{Bin}(n,\theta)$ the posterior mean $(k+\alpha)/(n+\alpha+\beta)$ trades a small bias for a smaller variance than the MLE k/n, and it wins whenever n is small and θ is not far from the prior mean $\alpha/(\alpha+\beta)$.

PART IV

Learning in AI, Agents and Games

The same counts, priors and EM everywhere

PART IV · AI CONNECTIONS

10.13 Connections to the Advanced AI Course

AI CONNECTION · Reinforcement Learning

Model-based RL (course session 20) learns transition CPTs by counting; posterior sampling for RL draws a model from the Dirichlet posterior (Theorem 10.4), plans in it, and acts. A robot executes "move right" from state s twenty times (hypothetical log) and lands in the goal 14 times, slips and stays 5 times, and slides back once: counts $(14,5,1)$ over $s' \in (\mathrm{goal}, \mathrm{stay}, \mathrm{back})$. The MLE transition row is $(14,5,1)/20 = (0.70, 0.25, 0.05)$; with a $\mathrm{Dir}(1,1,1)$ prior the posterior mean is $(15,6,2)/23 = (0.6522, 0.2609, 0.0870)$. With values $V(s') = (1,0,0)$ (only the goal is worth anything), the one-step expected value $\sum_{s'} \hat P(s' \mid s,a)V(s')$ is 0.7000 under the MLE and 0.6522 under the posterior mean. By Theorem 10.7 the estimation error decays as $1/\sqrt{n_{s,a}}$: in a simulation that draws n transitions from the row $(0.7, 0.25, 0.05)$ 2000 times (seed 2), the mean of the largest absolute error

$\max_{s'} |\hat{P}(s' \mid s,a) - P(s' \mid s,a)|$ is 0.0981, 0.0314, 0.0099 after $n = 20$, 200, 2000 visits, shrinking by about $\sqrt{10} = 3.16$ per tenfold increase.

PYTHON LAB · s10_examples.py · ex_model_based_rl()

```
MLE [0.7 0.25 0.05] -> E[V(s')] = 0.7000;  Dir(1) mean [0.6522 0.2609 0.087]
-> 0.6522; errors 0.0981, 0.0314, 0.0099.
```

LINK TO COURSE SESSION 18 · Exploration: Thompson sampling

Thompson sampling (Thompson, 1933) keeps a Beta posterior per arm (Theorem 10.2), samples one value per arm and pulls the arm with the largest sample, exploring as much as the posterior is uncertain. **Simulation** (`ex_thompson()`, `thompson_sim()`, seed 11): a Bernoulli bandit with three tools of success rates $(0.55, 0.65, 0.70)$, $T = 1000$ rounds, 300 independent runs, all agents starting from Beta$(1,1)$ priors. The *cumulative regret* after T rounds is $\sum_{t=1}^{T}(p^* - p_{a_t})$ with $p^* = 0.70$: the expected number of successes lost by not always calling the best tool. Its mean over the 300 runs is 59.79 for greedy (always call the tool with the largest Laplace-smoothed mean $(S+1)/(S+F+2)$, S, F = successes, failures), 25.54 for ε-greedy ($\varepsilon = 0.1$: a uniformly random tool with probability 0.1, otherwise greedy) and 19.16 for Thompson sampling (Russo et al. 2018).

PYTHON LAB · s10_examples.py · ex_thompson()

```
greedy :  mean cumulative regret at T=1000 = 59.79;        eps-greedy (0.1) :
25.54; Thompson :  19.16.
```

AI CONNECTION · Agentic AI

An agent that learns which tool to trust. An LLM assistant can call two search tools. With uniform Beta$(1,1)$ priors, tool A succeeded 18 times in 20 calls and tool B 4 times in 4: posteriors Beta$(19,3)$ (mean 0.8636, 95% interval $(0.6962, 0.9695)$) and Beta$(5,1)$ (mean 0.8333, interval $(0.4782, 0.9949)$). (The intervals are 95% equal-tailed credible intervals: the 2.5% and 97.5% Beta quantiles.) Thompson sampling draws θ_A, θ_B independently from the two posteriors and calls B iff $\theta_B > \theta_A$. Beta$(5,1)$ has cdf $P(\theta_B \le t) = t^5$, so conditioning on θ_A gives $P(\theta_B > \theta_A) = \mathbb{E}[1 - \theta_A^5]$; with the Beta$(a,b)$ moments $\mathbb{E}[\theta^m] = \prod_{i=0}^{m-1} \frac{a+i}{a+b+i}$,

$$P(\theta_B > \theta_A) = 1 - \frac{19 \cdot 20 \cdot 21 \cdot 22 \cdot 23}{22 \cdot 23 \cdot 24 \cdot 25 \cdot 26} = 1 - \frac{7980}{15600} = 1 - 0.5115 = 0.4885.$$

B has the lower mean but is still tried almost half of the time because it is so uncertain (a Monte Carlo check with 10^6 draws, seed 9, gives 0.4881); a greedy agent would always call A and never test B again.

PYTHON LAB · s10_examples.py · ex_tool_reliability()

```
E[theta_A^5] = 7980/15600 = 0.5115;        tool A: Beta(19,3) mean=0.8636 95%
CI=[0.6962 0.9695];     tool B: Beta(5,1) mean=0.8333 95% CI=[0.4782 0.9949];
P(theta_B > theta_A) exact = 0.4885, Monte Carlo = 0.4881.
```

AI CONNECTION · Multi-Agent AI

Fictitious play learns an opponent model by counting. In rock–paper–scissors (payoff $+1$ for a win, -1 for a loss, 0 for a tie), an agent has seen its opponent play R, P, S $12, 5, 3$ times. Its Dirichlet$(1,1,1)$ predictive for the opponent's next move is $(13, 6, 4)/23 = (0.5652, 0.2609, 0.1739)$. Expected payoffs: R beats S and loses to P, so $\mathbb{E}[\mathrm{R}] = 0.1739 - 0.2609 = -0.0870$; similarly $\mathbb{E}[\mathrm{P}] = 0.5652 - 0.1739 = 0.3913$ and $\mathbb{E}[\mathrm{S}] = 0.2609 - 0.5652 = -0.3043$. The best response is Paper. When two agents both play this way, each counting the other's moves with a Dir$(1,1,1)$ prior and best-responding to its predictive (fictitious play, Brown 1951; `fictitious_play_rps()`), the empirical frequencies of their moves converge to the Nash equilibrium $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ in this zero-sum game (Robinson, 1951); after 5000 rounds the second player's frequencies of R, P, S are $(0.328, 0.333, 0.339)$.

PYTHON LAB · s10_examples.py · ex_fictitious_play()

```
Dir(1) predictive = [0.5652 0.2609 0.1739];  expected payoff R,P,S = [-0.087
0.3913 -0.3043]; best response = P; FP frequencies = [0.328 0.333 0.339].
```

Treat the potential outcome $Y(1)$ (the outcome a unit would have under treatment, Session 15) as missing for untreated units: "MAR given covariates" is the ignorability assumption of Session 15, and complete-case bias is confounding bias.

PART IV · AI CONNECTIONS

10.14 Summary

Table 10.1. Cheat sheet of key formulas.

Object	Formula
Decomposition	$\ell(\theta) = \sum_i \sum_u \sum_x N[x,u] \log \theta_{x\|u}$
MLE	$\hat{\theta}_{x\|u} = N[x,u]/N[u]$
Dirichlet posterior	$\theta_{\cdot\|u} \mid \mathcal{D} \sim \mathrm{Dir}(\boldsymbol{\alpha}_u + \mathbf{N}_u)$; predictive $(\alpha_{x\|u} + N[x,u])/(\alpha_u + N[u])$
ELBO	$\log p(x;\theta) = \mathcal{L}(q,\theta) + \mathrm{KL}(q \Vert p(\cdot \mid x;\theta))$
GMM M-step	$\pi_k = N_k/N$, $\mu_k = \sum_j r_{jk} x_j / N_k$, $\sigma_k^2 = \sum_j r_{jk}(x_j - \mu_k)^2 / N_k$
Expected counts	$\bar{M}[x,u] = \sum_j P(x,u \mid \mathbf{o}^{(j)}; \theta^{(t)})$
Fisher / CR	$I_1 = 1/(\theta(1-\theta))$; $\mathrm{Var}(T) \geq g'(\theta)^2 / I_n(\theta)$
CPT asymptotics	$\sqrt{N}(\hat{\theta}_{x\|u} - \theta_{x\|u}) \xrightarrow{d} N(0, \theta(1-\theta)/p_u)$
Shrinkage MSE	$[n\theta(1-\theta) + (\alpha - (\alpha+\beta)\theta)^2]/(n+\alpha+\beta)^2$
EM rate	largest eigenvalue of $I_{\mathrm{com}}^{-1} I_{\mathrm{mis}}$

THINK ABOUT IT **An LLM is trained by maximum likelihood on internet text. Which of the missing-data mechanisms best describes the text that is *not* on the internet, and what does that imply for the model?**

THINK ABOUT IT **EM and a VAE both maximize an ELBO. What exactly is lost when the E-step is amortized by an encoder network?**

THINK ABOUT IT **A tool-using agent only logs a success label when the user clicks "thumbs up". Is the agent's Beta-posterior estimate of tool reliability still valid?**

Further reading. Koller and Friedman (2009), *Probabilistic Graphical Models*, Ch. 17 (parameter estimation) and Ch. 19 (partially observed data); Murphy (2022), Ch. 4; Bishop (2006), *Pattern Recognition and Machine Learning*, Ch. 9 (mixtures and EM); Casella and Berger (2002), Ch. 7 and 10 (MLE, Cramér–Rao, asymptotics); Little and Rubin (2002), *Statistical Analysis with Missing Data*; Dempster, Laird and Rubin (1977), *JRSS-B* 39; Sutton and Barto (2018), Ch. 2, 8.

CONNECTIONS AND TAKEAWAYS

We can now learn CPT parameters exactly when data is fully observed (a direct, decomposable payoff of the DAG factorization), and, with latent variables, up to a local maximum of the likelihood by alternating inference (Session 7) with the same decomposable MLE (EM). Two loose threads remain before this toolkit is genuinely practical. First, we have always assumed the graph structure G itself is known — but where does the graph come from when domain knowledge is incomplete? Second, the E-step's inference call assumed variable elimination is tractable, but Session 7 showed VE's cost explodes with treewidth — what happens when it is not tractable? **Session 11** addresses both: a brief look at learning structure from data (where the Dirichlet marginal likelihood of today's Part II becomes the Bayesian score), and a full treatment of sampling-based approximate inference (including Markov chain Monte Carlo) for exactly the cases where exact inference is computationally out of reach.